MIL-R-17772C(YD)
17 December 1982
SUPERSEDING
MIL-R-17772B(YD)
17 August 1972

MILITARY SPECIFICATION

RADIATOR, HEATING, BASEBOARD PANEL, STEAM AND HOT WATER

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a baseboard panel radiator of the combined radiation and convection heating type for use on hot water or 2-pipe steam system.

* 1.2 Classification. The baseboard panel radiators shall be of the following assembly sizes (lengths) as specified (see 6.2).

Size 1	Assembly length		1 foot
Size 1-1/2	"	"	1-1/2 feet
Size 2	"	"	2 feet
Size 2-1/2	"	"	2-1/2 feet
Size 3	"	"	3 feet
Size 3-1/2	"	"	3-1/2 feet
Size 4	"	"	4 feet
Size 4-1/2	"	"	4-1/2 feet
Size 5	"	"	5 feet
Size 5-1/2	"	"	5-1/2 feet
Size 6	"	"	6 feet

* 1.3 Definitive military specification part number. The definitive military specification part number for items described in this specification will be identified as shown in 6.5.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4520

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2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

THE HYDRONICS INSTITUTE (IBR)

IBR Ratings for Baseboard Radiation and Finned Tube (Commercial) Radiation.

(Application for copies should be addressed to the Hydronics Institute, 35 Russo Place, Berkeley Heights, NJ 07922.)

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form).
B1.2 - Pipe Threads (Except Dry Seal).
B2.1 - Dry Seal Pipe Threads.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The baseboard panel radiator shall be an assembly of cast iron or non-ferrous finned tube heating element with die-formed sheet metal enclosures. The radiator shall be for use with hot water or steam as specified (see 6.2), and shall be designed for installation along the base of walls or for replacing conventional baseboard molding. The length of each assembly shall be specified (see 6.2). The radiator shall include necessary component parts including panel accessories for a complete installation. Suitable fasteners shall be provided by the installing contractor.

* 3.2 First article. When specified (see 6.2), the contractor shall furnish a radiator for first article inspection and approval (see 4.3 and 6.4).

* 3.3 Certification. Prior to approval of the sample radiator unit furnished for first article inspection, if one is required, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the radiator unit he proposes to supply under this specification meets the requirements of the IBR Testing and Rating Code for Baseboard Type Radiation of The Hydronics Institute. Acceptable evidence of meeting this standard will be the IBR emblem and applicable data indicating compliance with the IBR rating program.

* 3.4 Standard commercial product. The unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

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* 3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

* 3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

* 3.7 Design. The radiator shall be designed to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment. The radiator shall withstand, without deformation, leakage, or damage a minimum pneumatic pressure of 45 pounds per square inch gage (psig), or a minimum hydrostatic pressure of 60 psig when tested as specified in 4.8.1. Screw threads for all thread securing or attaching devices and threaded parts shall be in conformance with ANSI B1.1, B1.2, or B2.1 as applicable.

3.8 Construction. The radiator shall be of the combined radiant and convector type with suitable openings at top and bottom of the assembly. Enclosure front and back covers shall be formed of not less than 22 gage steel sheets. Front covers shall be braced to back panels or be otherwise reinforced to insure a firm fit. Enclosure ends shall conceal core elements and pipe connections. A removable enclosure cover shall be provided for access to vents, valves, and pipe connections. The back of the cast iron section shall have finned surfaces cast integrally with the steam or hot water distribution section. The die-formed sheet metal enclosure shall have a nonferrous type of heating element constructed of copper tubing and aluminum or copper fins. The fins shall be uniformly spaced and mechanically bonded to the copper tubing. Sections shall be in such lengths as to provide for radiator assemblies in linear multiples of 1 foot, the shortest length to be not less than 2 feet. The overall height of each assembly shall not exceed 12 inches and depth shall not exceed 4 inches.

3.8.1 Dampers. When specified (see 6.2), full length dampers formed from sheet steel with provisions for adjustment in any position between fully opened and closed shall be provided. Dampers over 5 feet long shall be in multiple sections. When dampers provided are an integral component of the units enclosure, the dampers shall be constructed of not less than 20 gage steel. When dampers provided are within the enclosure, the dampers shall be constructed of not less than 20 gage steel.

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3.8.2 Connections. Push nipples connecting the cast iron sections shall not be less than 0.750 inch in diameter and connections shall be at both the top and bottom of the sections. The assembled sections shall be secured with tie bolts. Nonferrous heating elements shall have tubing sized for standard sweat solder type copper fittings for field joining.

3.8.3 Tappings. The end of cast iron sections shall have 0.750-inch tappings directly opposite the bottom nipple ports for supply and return connections. Air vent tappings at top of each end section shall be 0.125 inch.

3.9 Capacity rating. The heat emission steam or hot water capacity shall be as specified (see 6.2). The ratings shall be an approved IBR rating.

3.10 Resistance to water flow. The pressure drop per foot of cast iron heating elements, exclusive of entrance and exit losses, shall not exceed that of a 0.750-inch pipe 7.5 inches long. The pressure drop of a nonferrous type heating element shall be as shown in the IBR rating data.

* 3.11 Treatment and painting. Unless otherwise specified (see 6.2), the radiator shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the radiator, other than corrosion-resisting steel, shall be protected against corrosion and to present a neat appearance.

* 3.12 Workmanship.

* 3.12.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

* 3.12.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

* 3.12.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.12.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall

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be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

* 3.12.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. First article inspection shall be performed on one radiator unit when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.7 and the tests of 4.8. The first article may be either a first production item or a standard production item from the contractor's current inventory provided the radiator unit meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining radiator units to be furnished under the contract.

* 4.4 Quality conformance inspection. Quality conformance inspection shall be performed on the samples selected in accordance with 4.6. This inspection shall include the examination of 4.7, the tests of 4.8, and the requirements of 4.9.

* 4.5 Inspection lot. All units offered for delivery at one time shall be considered a lot for purposes of inspection. The sample unit shall be one complete radiator heater.

* 4.6 Sampling.

* 4.6.1 Sampling for examination. A random sample of radiators shall be

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selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level II. The Acceptable Quality Level (AQL) shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

* 4.6.2 Sampling for tests. A random sample of radiators shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level S-3. The AQL shall be 4.0 percent defective.

* 4.7 Examination. The first article, when a first article is required, and each sample selected in accordance with 4.6.1 shall be examined to verify compliance with this specification. Any unit in the sample containing one or more defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Examination shall be conducted as specified in table I.

TABLE I. Classification of defects.

Categories	Defects
Major:	
101	Holes or cracks in castings.
102	Material not suitable or not as specified.
103	Design not as specified.
104	Threads not as specified.
105	Like parts not interchangeable as specified.
106	Components missing or not as specified.
107	Treatment and painting not as specified.
108	Workmanship not as specified.
Minor:	
201	Identification marking not as specified, illegible, or missing.

* 4.8 Tests. The first article, when a first article is required, and each sample selected in accordance with 4.6.2 shall be tested. Any unit failing to pass the following tests shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Tests shall be conducted as outlined in the referenced documents as herein specified.

* 4.8.1 Operating pressure. Each radiator shall be subjected to either a pneumatic or a hydrostatic pressure test at the pressures specified in 3.7. The radiator shall be submerged in water if the pneumatic pressure test is used. Nonconformance to 3.7 shall constitute failure of this test. The test pressure shall be maintained for not less than 1 minute.

* 4.9 Packaging inspection. The preservation, packaging, packing, and marking of the units shall be examined to determine compliance with the requirements of section 5 of this specification.

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* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

* 5.1.1 Level A.

* 5.1.1.1 Preservation. Preservation specified herein shall conform to the applicable specifications listed and shall be applied in accordance with MIL-P-116.

* 5.1.1.2 Accessories. Uncoated ferrous surfaces of accessories shall be coated with type P-1 preservative in accordance with MIL-P-116.

5.1.1.3 Threaded and machined surfaces. Threaded and machined surfaces shall be preserved with coating with type P-1 preservative. Threaded openings shall be plugged with iron, steel, or plastic plugs.

5.1.1.4 Intermediate packaging.

* 5.1.1.4.1 Radiators. Radiators shall be shipped assembled in the specified installation lengths. Radiator assemblies shall be bundled in pairs with the finned surfaces back to back and separated by a fiberboard divider. Each pair of radiators shall be bound with not less than two flat steel straps duty, grade, finish, and seal type optional.

* 5.1.1.4.2 Accessories. The repair accessories shall be intermediate packaged in fiberboard boxes conforming to PPP-B-636, class weather-resistant. The contents shall be cushioned and braced to prevent movement.

* 5.1.2 Commercial. The radiators shall be preserved and packaged in a manner to prevent deterioration and damage during shipment.

* 5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

* 5.2.1 Level A. Radiators, fittings, and accessories packaged as specified in 5.1 shall be packed in accordance with the level A requirements of MIL-STD-794. The container shall be selected from table I of MIL-STD-794 for the applicable level of packing.

* 5.2.2 Level B. Radiators, fittings, and accessories packaged as specified in 5.1 shall be packed in accordance with the level B requirements of MIL-STD-794.

* 5.2.3 Commercial. The radiators shall be packed in a manner which will insure arrival at destination in satisfactory condition. Containers and packing shall comply with applicable carrier rules and regulations.

* 5.3 Marking. Radiators shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The baseboard heating radiators are intended for the heating of barracks, warehouses, dining halls, shops, and residential and office buildings.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Whether radiators will be used with hot water or steam and assembly lengths required (see 1.2, 3.1, and 6.5).
c. When first article inspection and approval is required (see 3.2).
d. When dampers shall be provided (see 3.8.1).
e. Steam or hot water capacity (see 3.9).
f. Treatment and painting, if other than manufacturer's standard practice (see 3.11).
g. Level of preservation and level of packing required (see 5.1 and 5.2).

* 6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be either a first production item consisting of one complete baseboard radiator or a standard production item from the contractor's current inventory as specified in 4.3. The contracting officer should include specific instructions in all acquisition instruments, regarding arrangement for examinations, tests, and approval of the first article.

* 6.5 Definitive military specification part number. The definitive military specification part number corresponds to the type, class, and other specific classification data applicable to radiator units covered by this specification, and defines the requirements of the options presented under this specification. The military specification number, the type and class code number, and the radiator units' other applicable code numbers are combined to form the definitive military specification part number.

* 6.5.1 Cataloging data. For cataloging purposes, the definitive military specification part numbers for the radiator units are assigned as follows:

R-17772C X X

Military Specification Number ----------
Material and Heating Medium ----------
Damper and Painting Options ----------

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* 6.5.2 Material and Heating Medium. The material of construction and the heating medium of the baseboard radiator assemblies are identified by a single number (see table II).

TABLE II. Code number to material and heat medium.

Heating medium	Material of construction	
	Cast iron	Sheet metal enclosure-fin tube
Steam	1	2
Hot water	3	4

* 6.5.3 Damper and painting options. The damper and painting options of the baseboard radiator assemblies are identified by a single number (see table III).

TABLE III. Code number to damper and painting options.

Damper option	Painting option	
	Manufacturer's standard	As specified in 6.2
With damper	1	2
Without damper	3	4

* 6.5.4 Assembly length (size). The length (size) of various baseboard assemblies are identified by a single alphabetic character (see table IV).

TABLE IV. Code number to length (size).

Size	Length	Code
1	1 foot	A
1-1/2	1-1/2 feet	B
2	2 feet	C
2-1/2	2-1/2 feet	D
3	3 feet	E
3-1/2	3-1/2 feet	F
4	4 feet	G
4-1/2	4-1/2 feet	H
5	5 feet	J
5-1/2	5-1/2 feet	K
6	6 feet	L

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-17772C(YD) - RADIATOR, HEATING, BASEBOARD PANEL, STEAM AND HOT WATER

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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Replaces edition of 1 Jan 72 which may be used.

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